

Eigenvalues, eigenvectors

Standard method:

- eigenvalues: roots of the characteristic polynomials.

Difficult problem!

- eigenvectors: solution of a singular set of linear equations:

Easy.

Power method: find the largest eigenvalue

Let $\underline{A}$ be an non-diagonalizable matrix with n eigenvalues such that $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|$ and with the corresponding eigenvectors $\underline{v}_i$. We know, that $\underline{v}_i$ span the whole space if $\underline{A}$ is diagonalizable.

④ Let us start from an arbitrary vector $\underline{b}_0 = c_1 \underline{v}_1 + c_2 \underline{v}_2 + \dots + c_n \underline{v}_n$, with $c_1 \neq 0$.

Apply the operator $\underline{A}$ to $\underline{b}_0$

$$\underline{A} \underline{b}_0 = \underline{A} \sum_i c_i \underline{v}_i = \sum_i c_i \underline{A} \underline{v}_i = \sum_i c_i \lambda_i \underline{v}_i$$

if k is sufficiently large then

$$|\lambda_1|^k \gg |\lambda_i|^k \quad i > 1$$

So $\lim_{n \rightarrow \infty} \underline{A}^n \underline{b}_0 = \lambda_1^k C_1 \underline{v}_1 = \alpha \cdot \underline{v}_1$ converges to the

eigenvector of the largest eigenvalue,

obviously we can get λ_1 by $\underline{A} \alpha \cdot \underline{v}_1 = \lambda_1 \cdot \alpha \cdot \underline{v}_1$

The practical method includes a norma-

lization:

$$\underline{b}^{(k+1)} = \frac{\underline{A} \underline{b}^{(k)}}{\|\underline{A} \underline{b}^{(k)}\|}$$

$\underline{b}_0$ has to be chosen carefully, or
two perpendicular starting points.

Eigenvalue shift

We can manipulate the eigenvalues of
the original matrix:

$$- \underline{A} \rightarrow \underline{A}^{-1} : \lambda_1, \dots, \lambda_n \rightarrow \frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_n}$$

$$- \underline{A} \rightarrow \underline{A} + \underline{I} \cdot \alpha : \lambda_1, \dots, \lambda_n \rightarrow \lambda_1 + \alpha, \dots, \lambda_n + \alpha$$

- Add a matrix with predefined
eigenvectors

We can manipulate the largest eigenvalue.

- General procedure: Shift close to the eigenvalue, then use the power method on the inverse. Useful if the approximate position is known

- Second largest eigenvalue with $\|v_1\| = 1$

$$\underline{A} \rightarrow \underline{A} - \lambda_1 \underline{v}_1 \underline{v}_1^T$$

$\underline{v}_1 \underline{v}_1^T = \underline{v}_1 \circ \underline{v}_1$ outer product has two

different eigenvalues $\lambda_1 = \|\underline{v}_1\|^2 = 1$ and

$$\lambda_2 = \dots = \lambda_n = 0$$

Numerical derivation

The definition of the derivative of the function $f(x)$ in the point a is

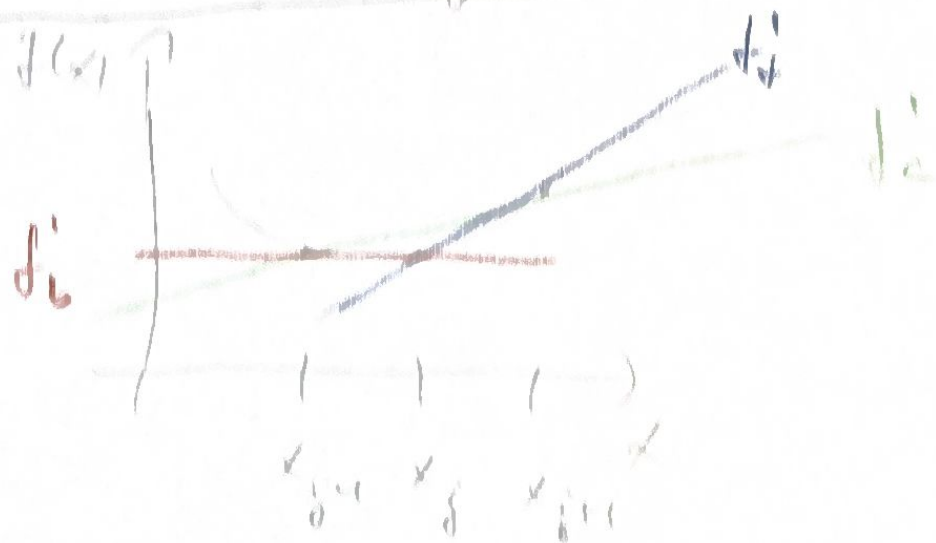
$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

Generally $f(x)$ is known at discrete points x_j $j = 1, 2, \dots, n$ $x_{j+1} - x_j = h$

Forward difference: $f'_f(x_j) = \frac{f(x_{j+1}) - f(x_j)}{x_{j+1} - x_j} = \frac{f(x_{j+1}) - f(x_j)}{h}$

Backward difference: $f'_b(x_j) = \frac{f(x_j) - f(x_{j-1})}{h}$

Central difference: $f'_c(x_j) = \frac{f(x_{j+1}) - f(x_{j-1}))}{2h}$



Error:

Start from the Taylor series

$$f(x) = \frac{f(x_j)(x-x_j)^0}{0!} + \frac{f'(x_j)(x-x_j)^1}{1!} + \frac{f''(x_j)(x-x_j)^2}{2!} + \dots$$

- Forward difference

$$x = x_{j+1}, \text{ again } x_{j+1} - x_j = h$$

$$f(x_{j+1}) = f(x_j) + f'(x_j)(x_{j+1} - x_j) + \frac{1}{2}f''(x_j)(x_{j+1} - x_j)^2 + \dots$$

$$f'(x_j) = \frac{f(x_{j+1}) - f(x_j)}{h} + \left(-\frac{1}{2}f''(x_j)h - O(h^2) \right)$$

$$f'(x_j) = \frac{f(x_{j+1}) - f(x_j)}{h} + O(h)$$

The error of the forward difference is h .

- Backward difference

Similarly the error is proportional to h .

Central difference: use the Taylor series at two points

$$f(x_{j+1}) = f(x_j) + f'(x_j)h + \frac{1}{2}f''(x_j)h^2 + O(h^3)$$

$$f(x_{j-1}) = f(x_j) - f'(x_j)h + \frac{1}{2}f''(x_j)h^2 + O(h^3)$$

$$f(x_{j+1}) - f(x_{j-1}) = 2f'(x_j)h + 0 + O(h^3)$$

$$f'(x_j) = \frac{f(x_{j+1}) - f(x_{j-1}))}{2h} + O(h^2)$$

One can go further and using more distant points cancel higher terms in the series. eg:

$$f'(x_j) = \frac{f(x_{j-2}) - 8f(x_{j-1}) + 8f(x_{j+1}) - f(x_{j+2}))}{12h} + O(h^4)$$

Important: numerical precision may limit the gain

